

3 **(long)** A cylindrical, open tank discharges an incompressible fluid of density ρ without friction to the atmosphere as shown in Fig. 3(a). The tank has diameter D and time-varying instantaneous water level height $h(t)$, which is initially $h(0) = H$. The discharge pipe has diameter d and negligible length, and the streamlines on exit can be assumed to be parallel. The acceleration of gravity is g .

- (a) Find the expression that relates the discharge velocity v with the rate of change of the water level, dh/dt . Under what conditions can the discharge be considered quasi-steady? [4]
- (b) Assuming that the flow is quasi-steady, find an expression for v in terms of D , d , g , and the instantaneous water level, h . [8]
- (c) Derive a differential equation for $h(t)$ and find the total discharge time T_D . To simplify your solution you may define a constant $G = \sqrt{\frac{gd^4}{2(D^4 - d^4)}}$. [10]
- (d) A length of pipe L with the same diameter d is added to the discharge outlet, as shown in Fig. 3(b). Find the new total discharge time, T'_D . Show that, for $L = H$, the new discharge time is $T'_D \approx 0.41 T_D$. [8]

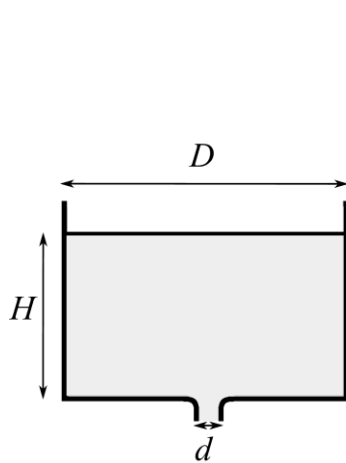


Fig. 3(a)

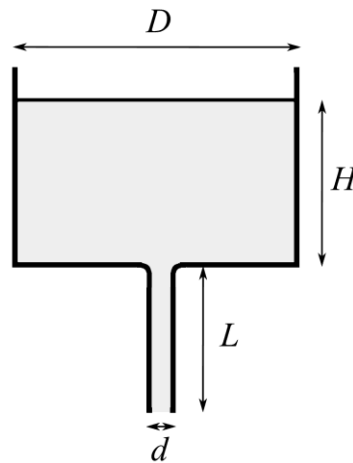


Fig. 3(b)